Family Name _____

First Name _____

Student Number |__|__|__|__|__|__|__|__|

Venue _____

Seat Number _____

UNIVERSITY OF CANTERBURY
Te Whare Wānanga o Waitaha
CHRISTCHURCH NEW ZEALAND

No electronic/communication devices are permitted.

No exam materials may be removed from the exam room.

Computer Science and Software Engineering

EXAMINATION

Mid-year Examinations

For Examiner Use Only

Question	Mark

Total _____

SENG201 (C) Software Engineering I

Examination Duration: 120 minutes

Exam Conditions:

Open Book exam: Students may bring in any written or printed materials.

Any scientific/graphics/basic calculator is permitted.

Materials Permitted in the Exam Venue:

Any non-electronic material (books, lecture notes, handwritten notes, printouts etc.) are allowed

Materials to be Supplied to Students:

1 x Write-on question paper/answer book

Instructions to Students:

No form of collaboration is permitted.

Answer all questions. In all answers be sure to provide clear and concise explanations of your reasoning. No marks will be given for answers not directly relevant to the question. Illegible handwriting and answers will not be marked. Answer all questions in the given space below each question.

This item of assessment is worth a total of 100 marks. Check carefully the number of marks allocated to each question. This suggests the degree of detail required in each answer, and the amount of time you should spend on the question.

Questions Start on Page 3

1. [8 marks for whole question] Do the following Java code fragments compile? If they do compile, what is their output? If they do not compile, explain why a fragment does not compile. Assume that fragments are taken from within a method of a Java class. (a) to (d) are not related (i.e., a variable in one question is independent from the same variable in another question).

   a. [2 marks]

```
int $villainEnergy = -1;
System.out.println($villainEnergy++ + " and " + $villainEnergy);
```

   Compiles: Yes/No (circle answer)
   Output (if statement compiles): _____
   Reason if statement does not compile: _____

   b. [2 marks]

```
String s1 = "SENG201";
String s2 = "SENG202";
s2 = s1;
String s3 = "SENG" + "201";
System.out.println(s3==s2);
```

   Compiles: Yes/No (circle answer)
   Output (if statement compiles): _____
   Reason if statement does not compile: _____

   c. [2 marks]

```
int up = 10;
int down = 4;
System.out.println(up/down);
```

   Compiles: Yes/No (circle answer)
   Output (if statement compiles): _____
   Reason if statement does not compile: _____

   d. [2 marks]

```
String switch = "OFF";
if (switch == "ON")
{
	System.out.println("The washing machine is on.");
}
else
{
	System.out.println("The washing machine is off.");
}
```

   Compiles: Yes/No (circle answer)
   Output (if statement compiles): _____
   Reason if statement does not compile: _____

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2. **[12 marks for whole question]** Study the following program. You can assume that classes `Lecturer`, `SerializeDemo` and `DeserializeDemo` are all implemented in files `Lecturer.java`, `SerializeDemo.java` and `DeserializeDemo.java` (line numbers are provided for your convenience; they are not part of the program).

```
 1  public class Lecturer implements java.io.Serializable {
 2          public String name;
 3          public String address;
 4          public static String nhi;
 5          public transient int ucnumber;
 6
 7  // You can assume that getters and setters are all implemented in this class
 8  }
```

```
 9  import java.io.*;
10
11  public class SerializeDemo {
12
13          public static void main(String[] args) {
14                  Lecturer e = new Lecturer();
15                  e.setName("Ernest Rutherford");
16                  e.setAddress("Brightwater, NZ");
17                  Lecturer.setNhi("ERR1871");
18                  e.setUcnumber(1);
19
20                  try {
21                          FileOutputStream fileOut = new FileOutputStream("lecturer.ser");
22                          ObjectOutputStream out = new ObjectOutputStream(fileOut);
23                          out.writeObject(e);
24                          out.close();
25                          fileOut.close();
26                          System.out.printf("Serialized data is saved in /lecturer.ser");
27                  } catch (IOException i) {
28                          i.printStackTrace();
29                  }
30          }
31  }
```

```
32  import java.io.*;
33
34  public class DeserializeDemo {
35          public static void main(String[] args) {
36                  Lecturer e = null;
37                  try {
38                          FileInputStream fileIn = new FileInputStream("lecturer.ser");
39                          ObjectInputStream in = new ObjectInputStream(fileIn);
40                          e = (Lecturer) in.readObject();
41                          in.close();
42                          fileIn.close();
43                  } catch (IOException i) {
44                          i.printStackTrace();
45                          return;
46                  } catch (ClassNotFoundException c) {
47                          System.out.println("Lecturer class not found");
48                          c.printStackTrace();
49                          return;
50                  }
51
52                  System.out.println("Deserialized Lecturer...");
53                  System.out.println("Name: " + e.getName());
54                  System.out.println("Address: " + e.getAddress());
55                  System.out.println("NHI: " + Lecturer.getNhi());
56                  System.out.println("UC Number: " + e.getUcnumber());
57          }
58  }
```

a. [6 marks] Explain in detail what happens in lines 21, 22 and 23.

Your answer:

b. [6 marks] `DeserializeDemo.java`, is a class that receives a `ser` file, deserializes its content and prints the state of the deserialized objects stored in the `ser` file. Suppose that the object defined in lines 14 to 18 is serialized and stored in `lecturer.ser`. Explain in detail what the output obtained from lines 52 to 56 is.

Your answer:

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3. **[20 marks for whole question]** Parts of this question relate to the UML class diagram shown in Figure 1. A SENG201 student has been asked to design a class framework suitable for representing vehicles. Figure 1 shows the latest version of the representation. The corresponding classes have been implemented in Java and all compile without error.

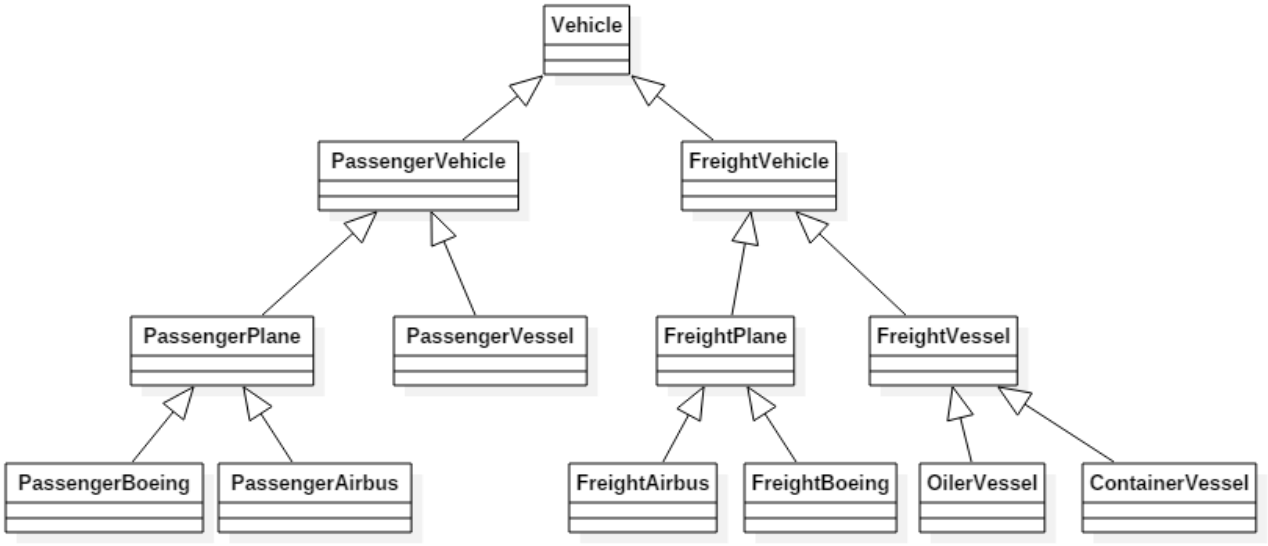


Figure 1: Vehicles UML class diagram

Variables are declared and initialised as follows:

```
1  Vehicle milleniumFalcon = new PassengerVehicle();
2  FreightVehicle nautilus = new FreightVehicle();
3  PassengerVehicle deLorean = new PassengerVehicle();
4  // an assignment goes here: see questions below
```

Line numbers are included for your convenience, they are not part of the program.

a. [15 marks] Which of the following assignments would then be legal replacements for the comment at line 4? For those which are legal, clearly and concisely explain their effect (i.e., what consequence does the assignment of the variables have?). For those which are not legal, clearly and concisely explain why they are not legal.

i. [3 marks]

```
milleniumFalcon = new PassengerAirbus();
```

Legal: Yes/No (circle answer)
Effect (if statement is legal): _____
_____
Reason (if statement is not legal): _____
_____

ii. [3 marks]

```
Vehicle gruMobile = new FreightPlane();
```

Legal: Yes/No (circle answer)
Effect (if statement is legal): _____
_____
Reason (if statement is not legal): _____
_____

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iii. [3 marks]

```
deLorean = new Vehicle();
```

Legal: Yes/No (circle answer)
Effect (if statement is legal): _____
_____
Reason (if statement is not legal): _____
_____

iv. [3 marks]

```
nautilus = deLorean;
```

Legal: Yes/No (circle answer)
Effect (if statement is legal): _____
_____
Reason (if statement is not legal): _____
_____

v. [3 marks]

```
milleniumFalcon = new Vehicle();
```

Legal: Yes/no (circle answer)
Effect (if statement is legal): _____
_____
Reason (if statement is not legal): _____
_____

b. [5 marks] Which classes in Figure 1 should be abstract classes, and what effect would the abstract class have on your answers to question 3.a?

Your answer:

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4. **[19 marks for whole question]** Figure 2 shows a class diagram for a "Heroes and Villains" game, similar to the game you developed for this year's SENG201 assignment. You can assume that all classes in the diagram have properties and methods necessary for the app to function correctly but were excluded from Figure 2 for simplicity.

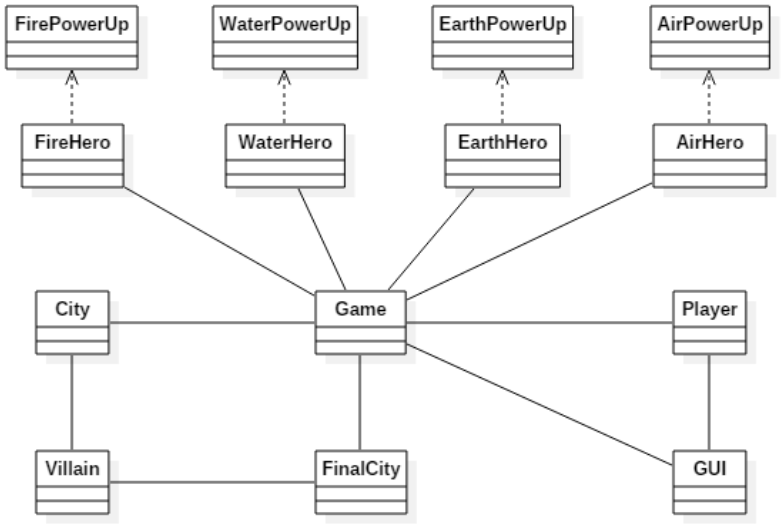


Figure 2. UML class diagram for Heroes and Villains game (omitting properties and methods)

Answer the following questions, and state all of the assumptions you make for your answers.

a. [9 marks] The class diagram shown in Figure 2 violates several good software engineering design principles. Identify and explain <u>two design principles</u> that are being violated in the class diagram.

Your answer:

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b. [10 marks] Propose and explain two design patterns to correct the violations identified in question 4.a.

Your answer:

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5. **[24 marks for whole question]** You are hired by a local software start-up to help develop a new app to control robot vacuum cleaners.
   a. [8 marks] Sketch a UML use case diagram to discuss the features of this new app with your colleagues. The app should:
      - Allow robot owners to sign up, start or pause cleaning cycles, schedule a routine cleaning cycle for any day of the week, select custom cleaning preferences, monitor activity of robots, check a list of places where robot cleaned, view statistics (about floor coverage, cleaning time, and charging time), and turn on/off real-time notifications.
      - Permit developers to update the software.
      - Allow tech support, for maintenance or repairing purposes, to activate special maintenance cleaning cycles.

      Your use case diagram should include at least three actors and at least six use cases, and must follow the notation of UML use case diagrams.

Your answer:

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b. [8 marks] Draw a state transition diagram for the life cycle of a vacuuming cycle. You can consider situations like overheat, filter clogged or power outage. The diagram should include at least three states. State all assumptions you make.

Your answer:

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c. [8 marks] Draft four unit test cases derived from the question 5.b diagram. Include pre and post-conditions of the test cases. Note that your test cases do not have to be fully formed JUnit tests.
Hint: An expected test case might look like this:

GIVEN: status = OFF //preconditions, the initial state
WHEN: power() //what caused the transition
THEN: status = ON //post-conditions, the expected state

Your answer:

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6. **[9 marks for whole question]** List and explain in bullet points three reasons for why software requirements specification phase can potentially reduce the rework and evolution costs throughout the lifetime of a software system.

Your answer:

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7. **[8 marks for whole question]** Version control systems allow developers to manage the changes to source code and other elements of the software configuration.
   a. [4 marks] Explain in your own words three different features that make version control systems particularly useful for software development.

Your answer:

   b. [4 marks] Briefly explain two types of files which you would not manage with a version control system.

Your answer:

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End of Examination